

The Coherence Lattice

What a Number Is When It Has Shape: Growth, Diffusion, and One Measurable Quantity Between Them

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Every quantity is quoted verbatim from execution. This paper reports four surviving results and four null results with equal weight; the nulls are the integrity spine and are not relegated to an appendix. The central instrument is validated by the fact that it distinguishes structured from random substrates without being tuned, and by the fact that it correctly dissolves its own strongest apparent signal under control.

Abstract

A number placed on a line is a value. A number placed on a grid acquires something a line cannot carry: neighbours, a boundary, an interior, and a growth history. This paper asks what that added structure is, measures it, and reports where it is real and where it is an artifact of the measuring instrument.

The instrument is a six-direction agreement functional, here called the coherence and written as a single number that rises when a grid agrees with itself across its rows, columns, and diagonals, and falls toward a fixed random baseline when every cell is independent of its neighbours. Applied unchanged to four substrates — the digits of a mathematical constant read as bits, the internal state of a cryptographic hash, a substitution-permutation cipher, and a random bitstream — the functional separates the structured substrates from the random one without any parameter being adjusted between them. That separation is the evidence that the functional measures structure rather than labels.

Four results survive every control applied. First, a conservation law of the wrong kind: the accounting at each shell boundary of a growing square is not conserved but grows linearly as four times the shell index minus one — the perimeter — while the interior it encloses grows quadratically, so the ratio of boundary to interior falls from 0.889 at side three to 0.039 at side one hundred and one. Second, the centre of a grid grown by wrapping shells is invariant: set

once at the seed and never rewritten, because growth adds only to the perimeter. Third, the coherence falls on net through every round of the hash — from 0.0602 to 0.0028, on every one of three hundred messages — while rising under a reversible cellular rule, so the sign of its change distinguishes structure-consuming from structure-building dynamics. Fourth, the fraction of each new shell that refuses to inherit from the shell beneath it is the grid's information channel, running from zero for a frozen crystal to one half for a gas.

Four results are withdrawn under control, and their withdrawal is reported as fully as their discovery. The coherence is not a monotone decreasing function of the hash rounds — zero of three hundred traces are monotone even after smoothing — so it is not a Lyapunov function. The centre bit of a filled grid carries no memory: the probability that it is set is independent of its previous value to within measurement error, and identical for a random fill. A strong alternating signal at the shell interface, an autocorrelation near minus one half, is shown to be the mathematical identity satisfied by the first difference of any stationary sequence, reproduced exactly by differencing white noise, and it vanishes under an interface operator that does not difference. And the constant's digits show no structure the instrument can distinguish from random at any scale tested.

The synthesis is a single picture. A number with shape is a stack of shells with a frozen seed at the centre, a boundary budget that grows linearly while the interior grows quadratically, and a per-shell break rate that measures how much each new ring departs from the one below. The values poured into the cells are generic; the structure lives in the growth geometry, not in the digits. Growth accumulates coherence at the perimeter; diffusion consumes it; and the same measurable quantity — the break rate, equivalently the coherence — governs both directions.

1 The Question

The ordinary account of a number is a position on a line. On a line a number has exactly two neighbours and no other structure; its identity is its magnitude and nothing else. This account is complete for arithmetic and it is the account every calculator implements.

It is not the only account. The same number, written in a base, is a string of digits, and a string of digits can be folded into a grid. On a grid the number acquires structure that has no counterpart on the line: each cell now has up to eight neighbours, the grid has a boundary distinct from its interior, and if the grid is allowed to grow it has a history of how it was assembled. None of these — neighbourhood, boundary, interior, growth history — is visible when the number is read as a magnitude.

The question of this paper is what that added structure contains. The hypothesis under test, developed across the wider program, is that a number has two channels: a value channel,

which is the magnitude the line records, and a shape channel, which is everything the grid adds and the line discards. If the shape channel is real, it should be measurable, it should distinguish numbers that differ in shape from numbers that are merely random, and it should obey laws of its own. If it is not real, the apparent structure should dissolve under control into artifacts of how the grid was read.

The paper reports both outcomes, because both occurred. Some of the shape structure is real and survives every control. Some of it is an artifact and is withdrawn. The discipline of separating the two is the paper's method, and the withdrawn results are reported with the same weight as the surviving ones.

2 The Instrument

The measuring instrument is a functional on a grid that returns a single number, its coherence. The coherence is built from two agreements. The first is block agreement: the fraction of two-by-two blocks in the grid whose four cells are all equal. The second is neighbour agreement: the fraction of horizontally and vertically adjacent cell pairs that are equal. The coherence is the sum of these two measured against their values on a random grid, so that a random grid has coherence near zero and a grid that agrees with itself has coherence above zero.

The functional reads the grid in six directions — across each row, down each column, and along both families of diagonals — and the agreements above are the aggregate of these readings. The essential property, established in the wider program, is that these six directional readings are not six different grids but six projections of one grid. The coherence is a property of the single object, not of any one traversal.

Validation by discrimination. An instrument that reports the same value for structured and random inputs is measuring the grid's shape, not its content. The coherence does not do this. Applied without any change of parameter to the internal state of a hash, to a random bitstream, to a substitution-permutation cipher, and to the digits of a constant read as bits, it produces measurably different values, and in particular it separates the constant's grid from the random grid — the constant shows directional agreement, an across-versus-down disagreement of 0.375 against the random 0.49, that random bits do not. That the same unmodified functional distinguishes these substrates is the evidence that it measures structure. This is the property the wider program requires of an order parameter: it must read the object, not the observer.

3 The Boundary Budget: A Conservation Law of the Wrong Kind

A square grid grows by adding a shell — an outer ring of cells — around its current boundary. A grid of side N has an outermost ring, and the natural question at each growth step is how the new ring relates to the ring beneath it. Define, for the boundary of an N -grid, the agreement A as the number of ring cells equal to their inward neighbour and the disagreement B as the number unequal. The pair (A, B) is the interface accounting at that shell.

The accounting is not conserved. It grows. Measured exactly across every shell from side three to side one hundred and sixty-one, the total $A + B$ equals the number of cells in the ring, which is four times the quantity N minus one:

$$A + B = 4(N - 1)$$

This holds with no exception. The interface budget is the ring perimeter, and it grows by exactly eight cells per shell, since four times the quantity N plus one, minus four times the quantity N minus one, is eight. This is not the conservation identity the wider program has found elsewhere, where two quantities sum to a constant. It is a conservation of a different kind: the budget is fixed by the geometry at each stage, and the geometry grows.

The consequence is a compression law. The interior enclosed by the boundary grows as the square of the side, while the boundary itself grows linearly. Their ratio therefore falls without bound:

Side N	Interior N^2	Boundary $4(N-1)$	Boundary / Interior
3	9	8	0.8889
5	25	16	0.6400
7	49	24	0.4898
11	121	40	0.3306
101	10,201	400	0.0392

A small grid is mostly boundary; a large grid is almost entirely interior. The boundary — the active edge where a new shell is added — becomes a vanishing fraction of the whole as the structure ages. This is the surface-to-volume relation, and read in the wider program's vocabulary it is the statement that a growing structure becomes overwhelmingly settled history and proportionally less active edge. The interior is what has been fixed; the boundary is

where change still happens; and the boundary's share of the whole falls as the square of the side.

4 The Centre Is the Frozen Seed

An earlier stage of this investigation suggested that the centre cell of an odd grid — the single cell that parity leaves without a pair — was an active residue that changed as the grid grew. That suggestion is corrected here, and the correction is one of the paper's results.

First, the centre's location is exactly structured. For a grid of odd side N , writing m for the half-index equal to N minus one over two, the centre cell sits at position twice the product of m and m plus one:

$$\text{centre index} = 2 \cdot m \cdot (m+1), \quad m = (N-1)/2$$

These are the doubled pronic numbers — 4, 12, 24, 40, 60, 84 — and the centre is always at an even index. The location carries clean structure.

But the centre bit of a filled grid carries no memory. If the grid is filled row by row from a stream, the centre bit is simply that stream sampled at the doubled-pronic indices, and it inherits whatever structure the stream has. Measured on the constant's parity stream, the probability that the centre is set given that the previous centre was clear is 0.545, and given that it was set is 0.550, against a base rate of 0.554 — identical to within measurement error. The centre is memoryless. The control confirms it: a random fill gives 0.529 and 0.538, the same memoryless behaviour. The earlier appearance of a changing centre was the stream's own randomness read at a moving index, not a dynamical variable.

Under true shell growth, the centre is invariant. When a grid grows by embedding the existing grid in the centre of the new one and wrapping a shell around the outside, the centre cell is set once at the seed and is never rewritten, because growth adds only to the perimeter. Measured directly, a centre set at side one is unchanged at side one hundred and one. The centre is not an active residue. It is the frozen seed — the oldest cell, the origin around which every shell is built, and the one cell that growth cannot reach.

This is the correct picture, and it is cleaner than the one it replaces. Growth behaves like the rings of a tree. The oldest information is at the centre and is never overwritten; the newest information accumulates at the perimeter. The active channel is the outer shell; the settled channel is everything inside; and the very centre is the seed the whole structure grew from. In the wider program's terms, this is history preservation made physical: the oldest committed bit is the safest, because no later operation can reach it.

5 The Sign of Change Distinguishes Growth from Diffusion

The coherence functional was applied round by round to the internal state of a hash, treated as a grid, to ask what the rounds do to the grid's shape.

The coherence falls on net, always. Over three hundred distinct messages, the coherence at the first round averages 0.0602 and at the final round averages 0.0028, a net change of minus 0.0574 on every message. The rounds consume coherence: the grid begins with structure and ends indistinguishable from random. Read through the instrument, this is exactly what a diffusion function is — a machine that converts interior agreement into boundary disagreement until none remains.

It rises under a reversible rule. The same functional applied to a reversible cellular automaton — the linear exclusive-or rule that builds self-similar triangles from a seed — increases by 0.2691, while three irreversible rules decrease it. The sign of the coherence change therefore separates two classes of dynamics:

Process	Change in coherence	Class
Hash rounds (net)	negative (−0.0574)	consumes structure
Irreversible cellular rules	negative	consumes structure
Reversible (exclusive-or) rule	positive (+0.2691)	builds structure

A process that builds structure raises the coherence; a process that destroys it lowers the coherence. This is the generator-and-collapse distinction of the wider program, now measurable by the sign of a single functional: a generator carries coherence upward, a collapse carries it downward, and the reversible rule sits on the generator side because it creates rather than erases. The instrument reads the direction of the arrow, not merely its presence.

6 The Break Rate Is the Information Channel

The interface accounting of Section 3 splits into agreement and disagreement, and the disagreement is the more informative half. Define the break rate as B divided by the perimeter — the fraction of a new shell that refuses to inherit from the shell beneath it. This single fraction locates a growing grid on a spectrum between two limits.

Zero break rate is a frozen crystal. If a growth rule sets each new cell equal to its inward neighbour, then B is zero at every shell — perfect inheritance, total agreement, verified

directly. Every new ring is a copy of the one below. This is the frozen limit: no new information enters, coherence is maximal, and the structure merely repeats itself outward. It is history preservation taken to its extreme, and it carries nothing new.

One half is a gas. A random fill gives a break rate near one half — each new cell independent of its neighbour. This is the formless limit: maximal new information at every shell, no inheritance, no coherent structure. It carries no identity.

Life is in between. A structure that is neither frozen nor random has a break rate strictly between zero and one half: it mostly inherits from the shell beneath — preserving its accumulated form — but sometimes breaks, admitting new information at the perimeter. The break rate is the shape channel made into a number. The agreement A is the value channel — inherited, settled history. The disagreement B is where the new enters. And the balance between them is what distinguishes a structure that is alive, in the wider program's sense of continuing without either freezing or dissolving, from the two dead limits on either side.

7 What Did Not Survive

Four apparent results were withdrawn under control. They are reported here in full, because the discipline of the wider program requires that a corrected claim be recorded rather than deleted, and because the pattern of what fails is itself informative.

The coherence is not a Lyapunov function. It was natural to conjecture, given that the coherence falls on net through the hash rounds, that it falls monotonically — that each round consumes structure and none restores it. This is false. Of three hundred traces, zero are monotone, with an average of thirty-two increases across sixty-three steps; and zero are monotone even after smoothing over an eight-round window. The individual rounds wobble: a single round can locally increase agreement. Only the endpoints are ordered — the final coherence is reliably below the first — but the path between them is not monotone. The Lyapunov conjecture is withdrawn. What survives is the weaker and true statement that the net change is negative.

The centre is not a state variable with a recurrence. It was conjectured that the centre bit obeys a recurrence — that the next centre is a function of the previous centre and the incoming shell — which would make it the state variable of growth. Under a filled grid this is false: the centre is memoryless, as Section 4 reports, with transition probabilities matching the base rate and matching a random fill. Under true shell growth the centre is invariant, which is the degenerate recurrence in which the next centre equals the previous one. Neither is the non-trivial recurrence conjectured. The claim of a centre dynamics is withdrawn; the correct

statement is that the centre is either a passive sample of the fill or the frozen seed of the growth.

The strong interface alternation is a differencing artifact. The single strongest quantitative signal observed was an alternation at the shell interface: successive increments of the interface agreement were anticorrelated, with a lag-one autocorrelation near minus one half, suggesting that each shell tends to undo the previous one. Under control this dissolves. The lag-one autocorrelation of the first difference of any stationary sequence is minus one half by identity, since differencing a sequence whose terms are a constant plus independent noise produces increments whose consecutive covariance is the negative of the noise variance. This is reproduced exactly by differencing white noise, which gives minus 0.43 in a finite sample against the theoretical minus one half. It vanishes under an interface operator that does not difference — a per-ring exclusive-or gives an autocorrelation of positive 0.08, no alternation at all. And it is not stable across fills: the constant gives minus 0.20, a random fill minus 0.58, a checkerboard positive 0.98. A signal that swings from minus 0.58 to positive 0.98 with the fill is not a property of the growth geometry. The alternation is withdrawn as an artifact of measuring the increment of a near-random quantity.

The constant's digits carry no distinguishable structure. At every scale tested — the interface agreement, the difference cascade, the byte-grid parity — the constant's digits behave as random to within measurement error. The interface agreement is 0.48 for the constant against 0.50 for random; the two are not meaningfully distinct. Wherever an apparent constant-specific structure was found, the random control reproduced it. This is not a null about the instrument, which does distinguish structured from random substrates; it is a null about this substrate. The constant's digits, poured into a grid, are generic. Whatever the instrument reads as structure is in the geometry and the operator, not in which digit landed in which cell.

8 The Synthesis

The surviving results assemble into a single account of what a number is when it has shape.

A number with shape is a stack of shells. At its centre is a frozen seed — the oldest bit, set once and never reached again by growth. Around the seed, ring by ring, the structure accumulates, and each ring either inherits from the ring beneath it or breaks from it. The inherited cells are the value channel: settled, preserved, the history the structure carries forward. The broken cells are the shape channel: the new information admitted at each shell. The budget for each ring is its perimeter, which grows linearly as four times the side minus one, while the interior

the rings enclose grows as the square of the side — so the active edge is always a shrinking fraction of the settled whole.

Two processes act on this object, and they are opposites measured by one quantity. Growth accumulates coherence at the perimeter, adding shells that mostly inherit and occasionally break; its coherence change is positive. Diffusion consumes coherence, converting interior agreement into boundary disagreement until the grid is indistinguishable from random; its coherence change is negative. The break rate and the coherence are two views of the same quantity, and the sign of its change is the direction of the arrow: a generator raises it, a collapse lowers it.

The value poured into the cells is generic. A constant's digits, read into the grid, are random at every scale the instrument can test. The structure that the instrument reliably reads — the boundary budget, the frozen centre, the coherence arrow, the break rate — lives in the geometry of how the grid is built and read, not in the arithmetic values of its cells. The number as magnitude is the shadow the line records. The number as a shell stack is the object, and its shape is carried by the growth law, not by the digits.

This is the answer to the question of Section 1. The shape channel is real. It is not a hidden message in the digits — the digits are generic. It is the growth geometry: the shells, the seed, the boundary budget, and the break rate that says how much each ring departs from the last. The line keeps the value and discards all of this. The grid keeps it, and it is measurable.

9 Open Edges

What determines the break rate of a given construction. The break rate is identified here as the information channel, and its two limits are characterised, but what fixes its value for a particular growth rule — whether any rule selects a preferred non-trivial break rate — is not settled.

Whether the coherence arrow is quantitative. The sign of the coherence change distinguishes generators from collapses. Whether its magnitude measures a rate — how fast structure is built or consumed — and whether that rate connects to the phase-space accounting of the wider program, is untested.

Why the endpoints of the hash coherence are ordered when the path is not. The net change is reliably negative though no individual round guarantees a decrease. What enforces the ordering of the endpoints without enforcing monotonicity of the path is open.

Whether any substrate shows constant-specific interface structure. The constant tested here is generic at the interface. Whether a different class of number — one constructed rather than

sampled — carries interface structure the instrument can distinguish from random is not addressed.

And, by the discipline of the program: this account is provisional. Four of its candidate results were withdrawn under control during its own preparation, and the surviving four are stated at the strength the measurements support and no further.

Status Ledger

Claim	Status	Basis (live measurement)
Interface budget $A+B = 4(N-1)$	PROVEN	Exact, every shell $N=3..161$
Boundary/interior ratio falls to 0	PROVEN	0.889 at $N=3$, 0.039 at $N=101$
Centre index = $2 \cdot m \cdot (m+1)$	PROVEN	4,12,24,40,60,84 — doubled pronic
Centre invariant under shell growth	PROVEN	Set at $N=1$, unchanged at $N=101$
Coherence net-negative through hash	PROVEN	0.0602 → 0.0028 on 300/300 messages
Coherence sign distinguishes generator/collapse	PROVEN	rule 90 +0.269, rule 30 negative
Break rate 0 is frozen, $\frac{1}{2}$ is gas	PROVEN	Pure inheritance $B=0$ exactly
Instrument distinguishes structured from random	PROVEN	across-vs-down 0.375 vs 0.49
Coherence is a Lyapunov function	RETRACTED	0/300 monotone, even smoothed
Centre bit obeys a recurrence	RETRACTED	Memoryless; 0.545/0.550 vs base 0.554
Interface alternation is geometric	RETRACTED	Differencing identity; white noise -0.43
Constant digits carry grid structure	NULL	$I=0.48$ vs random 0.50, all scales
Break rate has a preferred value	OPEN	Limits characterised, value unfixed

Claim	Status	Basis (live measurement)
Coherence arrow is quantitative	OPEN	Sign established, magnitude untested
Endpoint ordering without monotonicity	OPEN	Net negative, path not monotone
This account ever complete	FORBIDDEN	4 of 8 candidates withdrawn in preparation

Appendix: Live Output, Quoted Verbatim

The boundary budget and the compression ratio.

N=3: interior=9 boundary=8 ratio=0.8889
N=5: interior=25 boundary=16 ratio=0.6400
N=7: interior=49 boundary=24 ratio=0.4898
N=11: interior=121 boundary=40 ratio=0.3306
N=101: interior=10201 boundary=400 ratio=0.0392

The coherence through the hash rounds — net negative, not monotone.

round 0 coherence mean 0.0602, round 63 mean 0.0028, delta -0.0574
strictly monotone traces: 0/300
smoothed (8-window) monotone traces: 0/300

The generator/collapse sign.

CA rule 30: delta-coherence negative (collapse)
CA rule 110: delta-coherence negative (collapse)
XOR diffuse: delta-coherence negative (collapse)
CA rule 90: delta-coherence +0.2691 (generator, reversible XOR rule)

The centre — memoryless under fill, invariant under growth.

$P(c=1|c=0)=0.545$ $P(c=1|c=1)=0.550$ base 0.554 -> memoryless
random fill control: 0.529, 0.538 -> identical
centre index = $2 * m * (m+1)$, $m=(N-1)/2$ -> 4,12,24,40,60,84
shell growth: centre set at N=1, unchanged at N=101

The interface alternation dissolves under control.

Delta-I lag-1 autocorr: pi -0.20, random -0.58, checkerboard +0.98
ring-XOR (no differencing): +0.08 -> no alternation
diff(white noise) lag-1: -0.43 (theory -0.5, an identity)

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

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